

Data quality issues and profiling of a car park fines dataset

May 31, 2026

1 Introduction

This report has been produced by applying the vizdataquality package to investigate data quality of a dataset about fines that were issued for all of the car parks in a city, during a three month period. The data is (c) Leeds City Council, 2019, <https://datamillnorth.org/dataset/v8ggw/off-street-parking-fines>, and is licensed under the terms of the Open Government Licence (<https://www.nationalarchives.gov.uk/doc/open-government-licence/version/2/>).

The first five workflow steps investigate data quality and clean it in a structured manner. The 6th step profiled the cleaned data.

2 Step 1: Is anything obviously wrong (look at your data and any documentation)?

Each row of the data file ends with a comma, which produces an empty, unnamed column. That column has been removed in this step.

3 Step 2: Watch out for special values

Sometimes datasets use special values (e.g., large integer values like 999, or old dates like 1 January 1800) to indicate values that are missing, invalid, etc. No such special values were found in this dataset.

4 Step 3: Is any data missing?

None of the variables are missing any values.

5 Step 4: Check each variable

A plot of each variable's value lengths revealed issues with LOCATION and CONTRAVENTION. Detailed investigation showed that some LOCATION and CONTRAVENTION values contain trailing spaces, so the data has been cleaned by stripping those trailing spaces.

The two date variables (ISSUED and Last Pay Date) only contain year, month and day (not time).

The PCN (the ID for a penalty charge notice) is not unique. Each PCN appears 1 to 5 times.

There are two values of FINE, but an unexpected number of different Balance and Total Paid values.

6 Step 5: Check combinations of variables

It would be logical to expect $\text{FINE} - \text{Total Paid} = \text{Balance}$. However, that is only true for 785 (11

7 Step 6: Profile the cleaned data

Table 1 shows the number of rows and columns in the dataset.

Table 1: The size of the dataset.

Dataset	Number of rows	Number of columns
../examples/Quarter 4 201819.csv	7108	8

Table 2 shows information about each variable.

Table 2: Data types and example values.

Variable	Data type	Example value
PCN	object	LS04961495
ISSUED	datetime64[ns]	2019-02-14 00:00:00
LOCATION	object	WOODHOUSE LANE CP - CITY
CONTRAVENTION	object	83 WITHOUT DISPLAYING A VALID TICKET
FINE	int64	50
Last Pay Date	datetime64[ns]	2019-03-28 00:00:00
Total Paid	float64	25.000000
Balance	float64	0.000000

Figure 1 shows that every variable is complete, but none are unique for every record.

Figure 2 shows that 1430 PCNs occur once, and the other 2511 PCNs each occur 2 - 5 times.

Figure 3 shows that some car parks have only one or two records, whereas one car park has more than 1000.

Figure 4 shows that Contravention 83 occurs in more records than all of the other contraventions put together, some of which only occur once or twice.

Figure 5 shows that most issued dates are from the past two years, but a few are from 10 years ago.

Figure 6 shows that, as expected, all of the Last Pay Dates are in Jan - Mar 2019.

Figure 7 shows that FINE is 50 or 70 pounds, but there are many unexpected values of Total Paid and Balance.

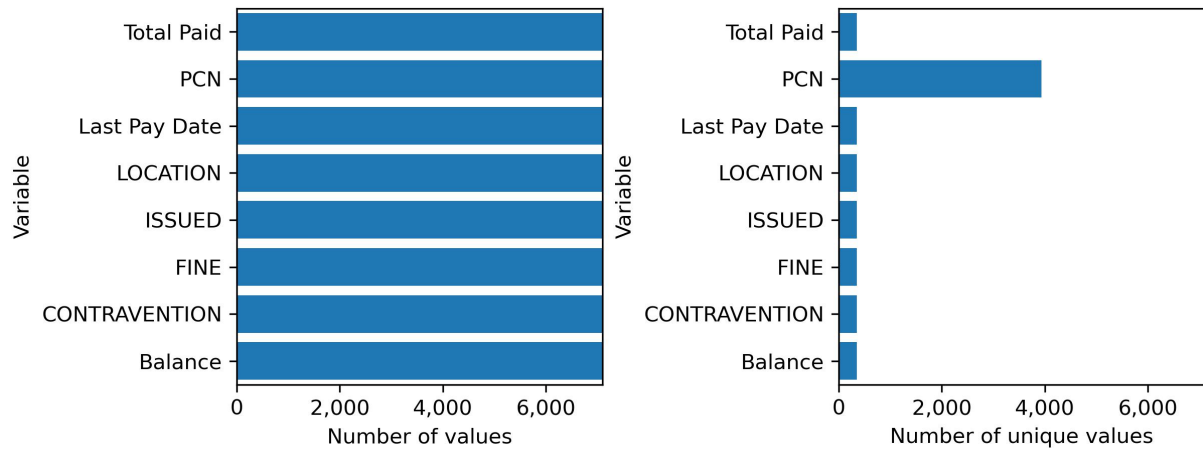


Figure 1: The number of values and unique values in each variable.

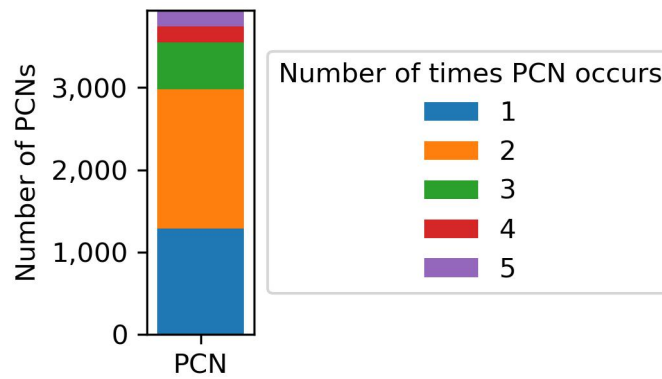


Figure 2: Number of occurrences of PCNs.

Table 3 shows that only for fines paid during the normal time period (no discount or penalty) is the following rule true: $\text{Balance} = \text{FINE} - \text{Total Paid}$.

Table 3: Relationship between Balance, FINE, Total Paid and the difference between the ISSUED and Last Pay Date.

FINE, Total Paid	Balance	Balance	0-14 days	15-56 days	57-70 days	71+ days
Consistent	Negative	0	0	0	0	0
Consistent	Zero	0	892	0	0	0
Consistent	Positive	0	6	0	0	0
Inconsistent	Negative	0	0	1	6	
Inconsistent	Zero	4753	0	254	693	
Inconsistent	Positive	328	0	62	113	

8 Acknowledgements

This report was created using the vizdataquality Python package.

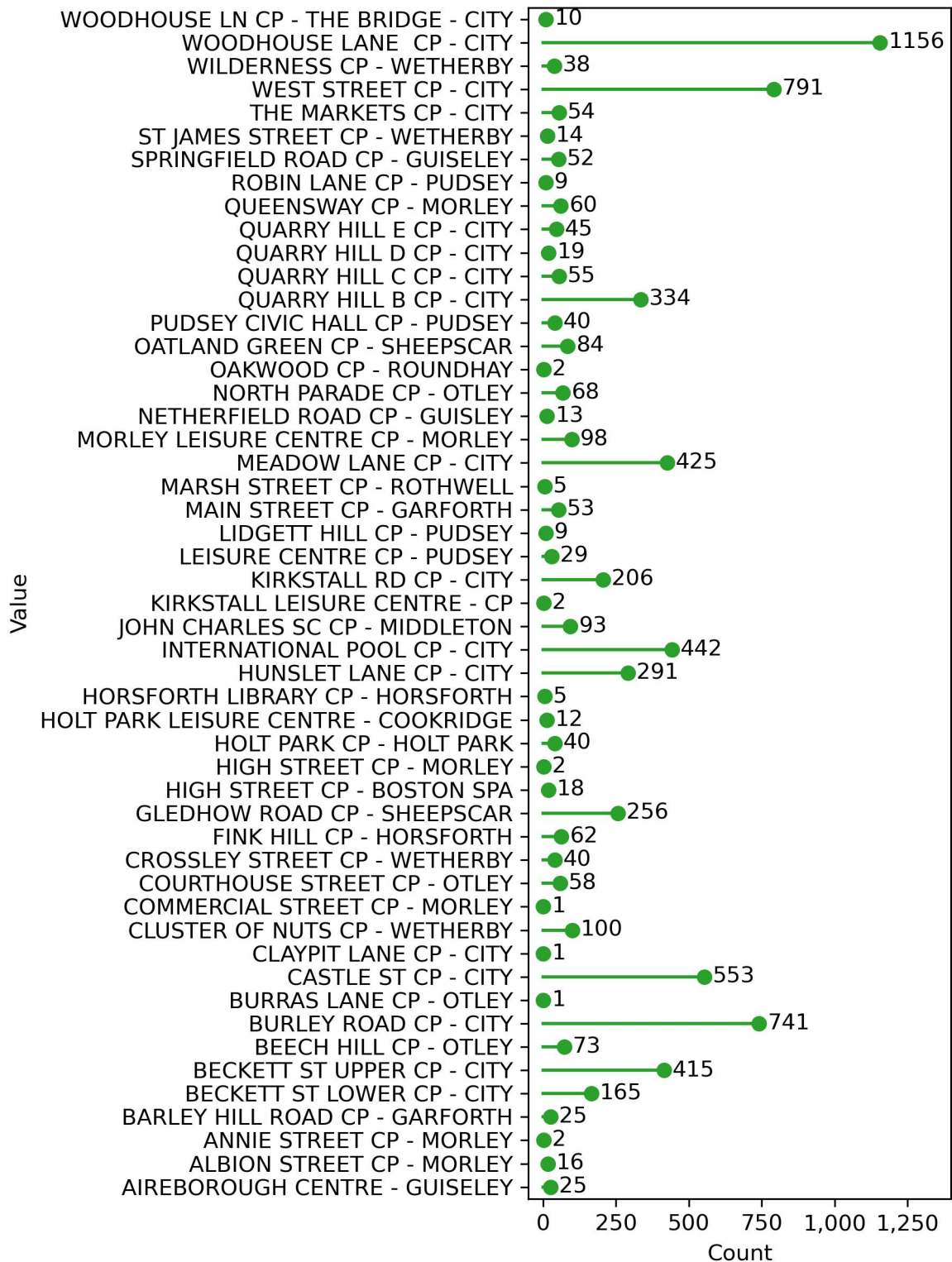


Figure 3: Number of records for each LOCATION.

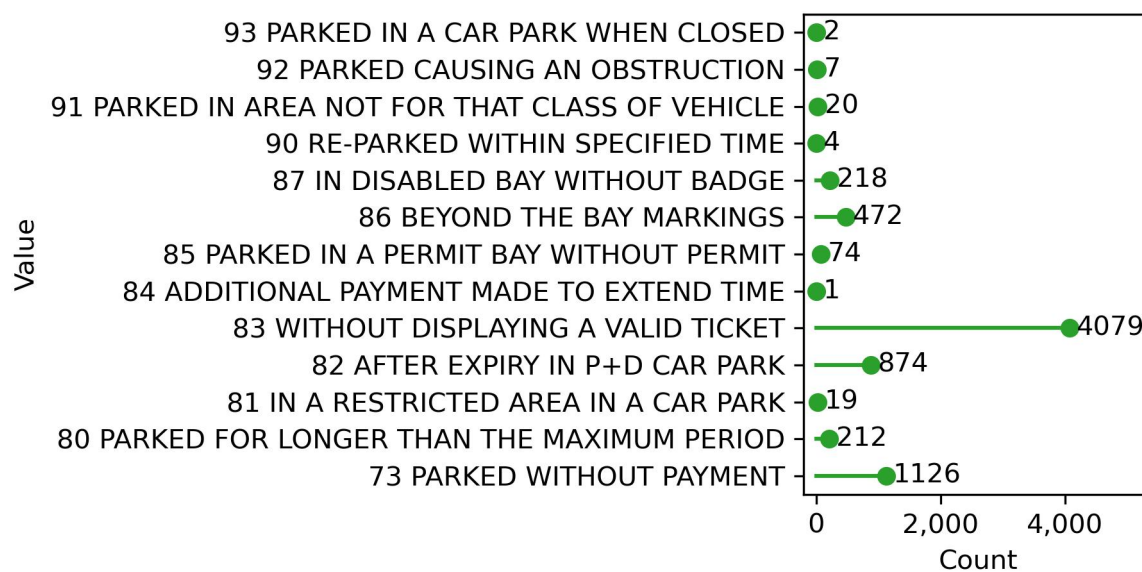


Figure 4: Number of records for each CONTRAVENTION.

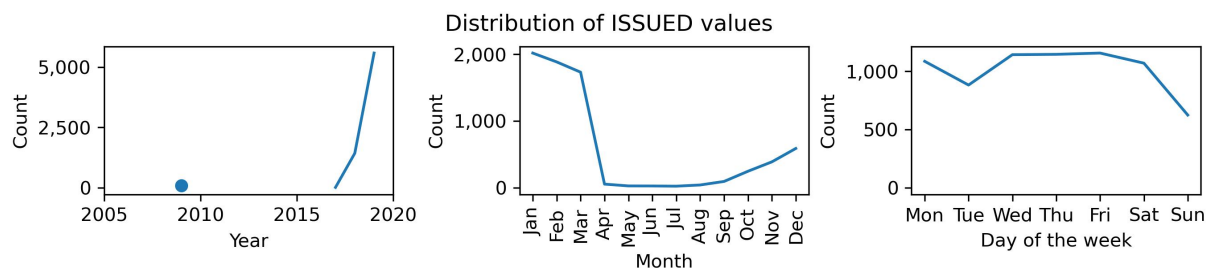


Figure 5: Distribution of ISSUED dates.

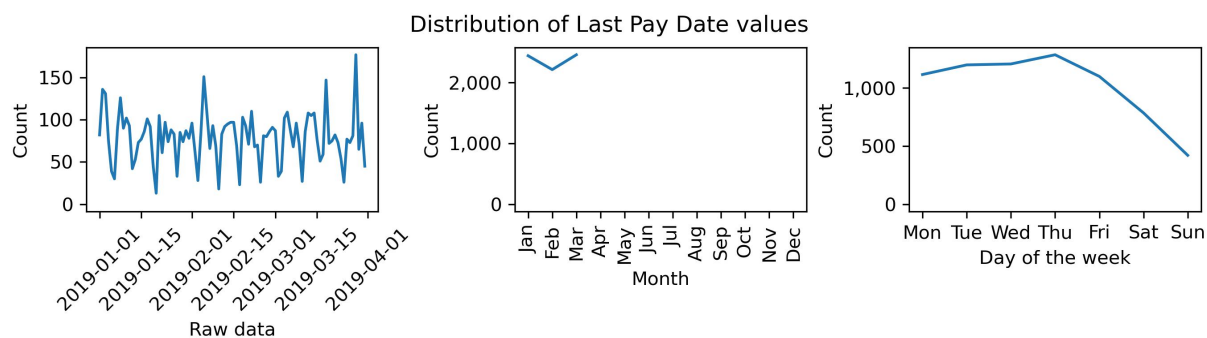


Figure 6: Distribution of Last Pay Date dates.

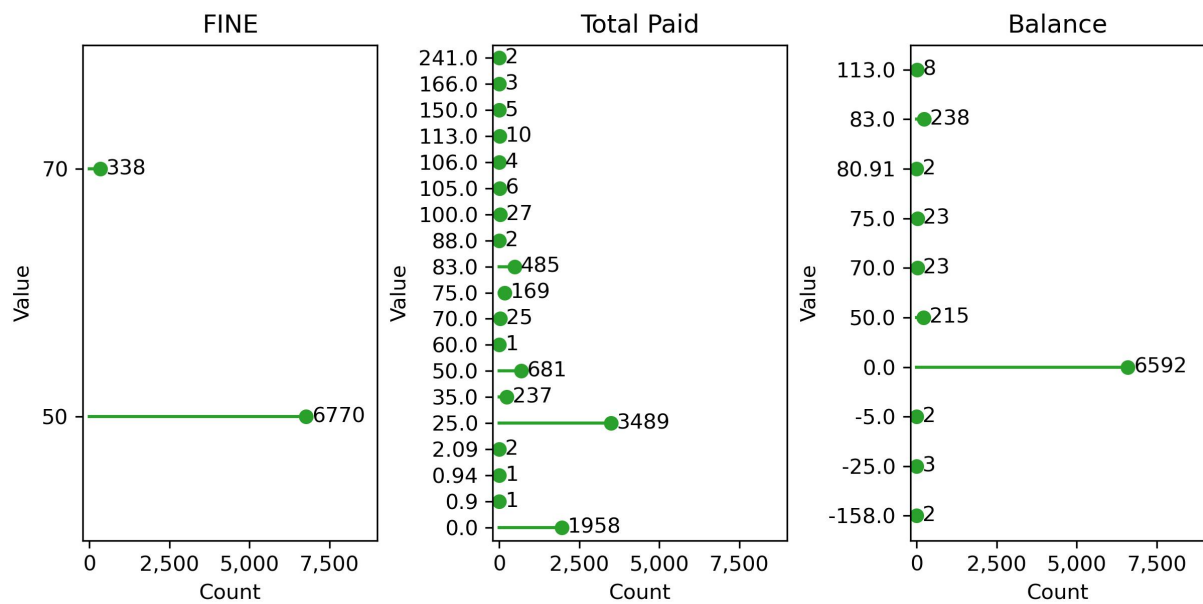


Figure 7: Value counts for the fine, total paid and balance.